Hints: Continuous functions – 1

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Answer: no. Look for a function for which multiplying by c is the same as a horizontal shift.

Hint 2. $f(x) = e^x$: $cf(x) = e^{x+\ln c}$, so the graph of cf is the graph of f translated by $-\ln c$ horizontally.

Problem 2

Hint 1. At a rational p/q: $f(p/q) = \frac{1}{q} > 0$, but there are irrational points arbitrarily close where $f = 0$.

Hint 2. At an irrational x and for $\varepsilon > 0$: in $(x-1, x+1)$ there are only finitely many fractions p/q with $q \leqslant \frac{1}{\varepsilon}$. Choose δ so small that none of them lies in $(x-\delta, x+\delta)$.

Hint 3. Second half: the set of continuity points of any f is $\bigcap_n \{x : \operatorname{osc} f(x) < \frac{1}{n}\}$, an intersection of countably many open sets. Show that $\mathbb{Q}$ is not such an intersection.

Hint 4. If $\mathbb{Q} = \bigcap U_n$ with U_n open (hence dense), then the countably many open dense sets U_n and $\mathbb{R} \setminus \{q\}$, $q \in \mathbb{Q}$, have empty intersection, which contradicts the Baire category theorem.

Problem 3

(a)

Hint 1. Answer: yes, with no continuity needed. Consider $A = \{x \in [0,1] : f(x) \geqslant x\}$ (it contains 0) and $s = \sup A$.

Hint 2. For $x \in A$: $x \leqslant f(x) \leqslant f(s)$, hence $s \leqslant f(s)$. Then $f(f(s)) \geqslant f(s)$, i.e. $f(s) \in A$, so $f(s) \leqslant s$.

(b)

Hint 1. Answer: no. Take a decreasing function that jumps over the diagonal, e.g. $f = 1$ on $[0, \frac{1}{2})$ and $f = 0$ on $[\frac{1}{2}, 1]$.

Problem 4

Hint 1. Fix an interval $(a, b) \subseteq [0,1]$. It suffices to find $p < q < r$ in (a, b) with $f(q) < \min(f(p), f(r))$: then the minimum of f on $[p, r]$ is attained at an interior point, which is a local minimum.

Hint 2. Suppose no such triple exists, i.e. $f(q) \geqslant \min(f(p), f(r))$ for all $p < q < r$ in (a, b). Take $[c, d] \subset (a, b)$ and a maximum point m of f on $[c, d]$.

Hint 3. For $c \leqslant s < t \leqslant m$ we get $f(t) \geqslant \min(f(s), f(m)) = f(s)$, so f is non-decreasing on $[c, m]$; similarly it is non-increasing on $[m, d]$. One of these intervals is non-degenerate, which contradicts the assumption that f is nowhere monotone.

Problem 5

Hint 1. If f is constant, take $X = [0, 1]$. Otherwise pick p, q with $f(p) \neq f(q)$, say $p < q$ and $f(p) < f(q)$, and for every $y \in (f(p), f(q))$ choose a preimage in a canonical way.

Hint 2. Let $x(y) = \min\{x \in [p, q] : f(x) = y\}$ (the first time f reaches y; it exists by the IVT and closedness). For $y_1 < y_2$, f takes the value y_1 on $[p, x(y_2)]$, so $x(y_1) < x(y_2)$.

Hint 3. Thus $X = \{x(y)\}$ has the cardinality of the continuum and f is strictly increasing on X.

Problem 6

Hint 1. Restate the hypothesis: if $f(z_n)$ converges, then z_n converges. Apply this to sequences that alternate between two points.

Hint 2. Alternating between two preimages of the same value shows that f is injective, hence a bijection. Interleaving z_n with a fixed z shows: $f(z_n) \to f(z)$ implies $z_n \to z$. So f^{-1} is continuous.

Hint 3. $f^{-1} : \mathbb{R} \to \mathbb{R}$ is a continuous bijection, hence strictly monotone (Fact 1), and then so is f. A strictly monotone map of $\mathbb{R}$ onto $\mathbb{R}$ has no jumps, so f is continuous.

Problem 7

Hint 1. For fixed x put $a_n = f(nx)$; then $|a_{m+n} - a_m - a_n| \leqslant M$. By induction $|a_{mn} - n\, a_m| \leqslant (n-1)M$.

Hint 2. Hence $\left|\frac{a_{mn}}{mn} - \frac{a_m}{m}\right| \leqslant \frac{M}{m}$ and, symmetrically, $\leqslant \frac{M}{n}$ with a_n/n. So $\left|\frac{a_m}{m} - \frac{a_n}{n}\right| \leqslant \frac{M}{m} + \frac{M}{n}$: the sequence is Cauchy.

Hint 3. The limit g satisfies $|g(x) - f(x)| \leqslant M$ and $g(x + y) = g(x) + g(y)$. As f is bounded near 0, so is g, say by K on $(-\delta, \delta)$; then $|g(x)| = |g(nx)|/n \leqslant K/n$ for $|x| < \delta/n$, which gives continuity at 0.

Hint 4. An additive function continuous at 0 is $g(x) = cx$. Then $|f(x) - cx| \leqslant M$, so $f(x)/x \to c$.

Problem 8

Hint 1. Put $h = f \circ g^{-1}$. It is a continuous bijection with $h + h^{-1} = 2\,\mathrm{id}$, and $y_0 = f(x_0)$ is a fixed point of h. You need $h = \mathrm{id}$.

Hint 2. h is strictly increasing (if it were decreasing, $h + h^{-1}$ would be decreasing). Substituting $h(y)$ into $h^{-1}(z) = 2z - h(z)$ gives $h(h(y)) - h(y) = h(y) - y$.

Hint 3. So the orbit $h^n(y) = y + n\, d(y)$, $n \in \mathbb{Z}$, is an arithmetic progression with $d(y) = h(y) - y$. If $y > y_0$ and $d(y) \neq 0$, some $h^n(y)$ is $< y_0$, while monotonicity gives $h^n(y) > h^n(y_0) = y_0$. The case $y < y_0$ is the same.

Problem 9

Hint 1. Show that $f(b) = \max_{[a,b]} f$ for every $a < b$; this gives $f(a) \leqslant f(b)$.

Hint 2. Let $M = \max_{[a,b]} f$ and let c be the largest point of $[a, b]$ with $f(c) = M$. If $c < b$, apply the hypothesis to $[c, b]$: there the only maximum point is c, so a minimum point $c_1 \leqslant c$ must equal c.

Hint 3. Then $\min_{[c,b]} f = \max_{[c,b]} f$, so f is constant on $[c, b]$ and $f(b) = M$.

Problem 10

Hint 1. $f \equiv 1$ works (with $a \geqslant 1$). Suppose $f(x_0) > 1$ for some x_0; by continuity you may take $x_0 \neq 0$ and $f \geqslant 1 + \varepsilon$ on $(x_0 - \delta, x_0 + \delta)$.

Hint 2. Choose $x = x_0/m$ and $n = 2m$. Every factor is $\geqslant 1$, and the factors with $jx \in (x_0 - \delta, x_0 + \delta)$, i.e. $|j - m| < m\delta/|x_0|$, are $\geqslant 1 + \varepsilon$. There are at least cn of them for some constant $c > 0$.

Hint 3. So the product is at least $(1 + \varepsilon)^{cn}$, which eventually exceeds an^k. Answer: only $f \equiv 1$.